

Name: **SOLUTIONS**

GCSE (1 - 9)

Area of a Sector



Buy me a coffee

Solutions by

Kenneth Stafford
MATHS TUTOR

ken@kennethstafford.co.uk

These questions are compiled from *released* [Pearson Edexcel GCSE maths exam papers](#)

They contain exam questions from up until the **Summer 2023** exam series

More recent exam papers do exist but are withheld for mock exams

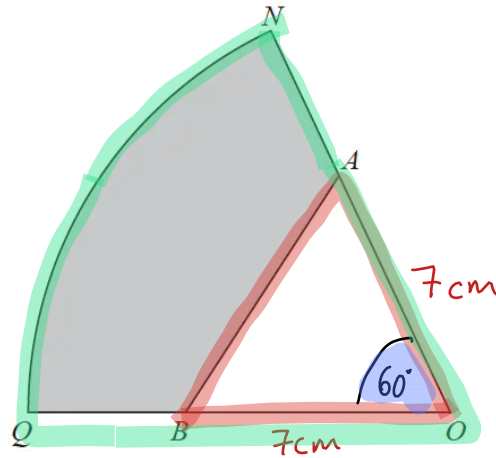
As new exam papers are released these worksheets will gradually be updated

The **most current** version of these worksheets will always be available at: kennethstafford.co.uk

Question	Where the question appeared
1	Edexcel, GCSE, Summer 2017, Paper 2H, Q17
2	Edexcel, GCSE, Autumn 2017, Paper 2H, Q20
3	Edexcel, GCSE, Autumn 2018, Paper 1H, Q17
4	Edexcel, GCSE, Summer 2019, Paper 2H, Q12
5	Edexcel, GCSE, Autumn 2019, Paper 3H, Q23
6	Edexcel, GCSE, Autumn 2020, Paper 1H, Q22
7	Edexcel, GCSE, November 2022, Paper 2F, Q21

Area of a Sector Questions from Past Papers

1



ONQ is a sector of a circle with centre O and radius 11 cm.

A is the point on ON and B is the point on OQ such that AOB is an equilateral triangle of side 7 cm.

Calculate the area of the shaded region as a percentage of the area of the sector ONQ .
Give your answer correct to 1 decimal place.

$$\text{Area of } AOB = \frac{1}{2} \times 7 \times 7 \times \sin 60 = 21.2176 \text{ cm}^2$$

$$\text{Area of sector } ONQ = \frac{60}{360} \times 11^2 \times \pi = 63.3555 \text{ cm}^2$$

$$63.3555 - 21.2176 = 42.1378$$

$$\frac{42.1378}{63.3555} \times 100 = 66.510$$

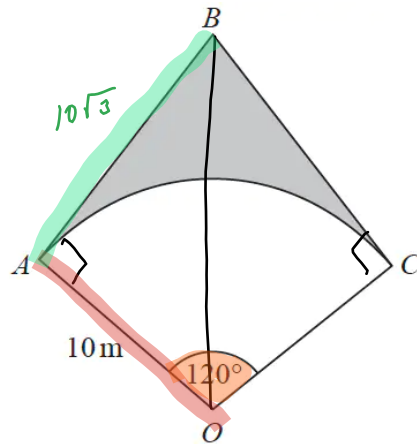
66.5 %

(Total for Question 17 is 5 marks)

[Edexcel, GCSE, Summer 2017, Paper 2H, Q17]

Area of a Sector Questions from Past Papers

2



OAC is a sector of a circle, centre O , radius 10 m .

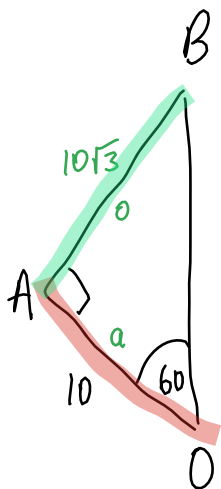
BA is the tangent to the circle at point A .

BC is the tangent to the circle at point C .

Angle $AOC = 120^\circ$

Calculate the area of the shaded region.

Give your answer correct to 3 significant figures.



$$AB: \text{opp} = 10 \times \tan(60^\circ) = 10\sqrt{3}$$

$$\begin{aligned} \text{Area of } \triangle AOB &= \frac{1}{2} \times 10 \times 10\sqrt{3} \\ &= 50\sqrt{3} \end{aligned}$$

$$\text{Area of } ABCO = 2 \times 50\sqrt{3} = 100\sqrt{3}$$

$$\begin{aligned} \text{Area of } \text{sector } AOC &= \frac{120}{360} \times 10^2 \times \pi \\ &= 104.7197 \end{aligned}$$

$$\begin{aligned} \text{Shaded region} &= 100\sqrt{3} - 104.7197 \\ &= 68.5 \text{ m}^2 \end{aligned}$$

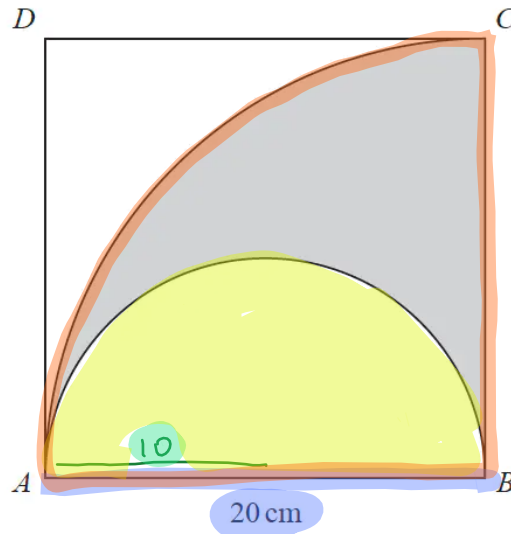
(Total for Question 20 is 5 marks)

[Edexcel, GCSE, Autumn 2017, Paper 2H, Q20]

Area of a Sector Questions from Past Papers

3

The diagram shows a square $ABCD$ with sides of length 20 cm. It also shows a semicircle and an arc of a circle.



AB is the diameter of the semicircle.
 AC is an arc of a circle with centre B .

Show that $\frac{\text{area of shaded region}}{\text{area of square}} = \frac{\pi}{8}$

$$\text{Area of square} = 20 \times 20 = 400 \text{ cm}^2$$

$$\text{Area of semicircle} = \frac{1}{2} \times 10^2 \times \pi = 50\pi \text{ cm}^2$$

$$\text{Area of sector} = \frac{1}{4} \times 20^2 \times \pi = 100\pi \text{ cm}^2$$

$$\text{Area of shaded region} = 100\pi - 50\pi = 50\pi \text{ cm}^2$$

$$\frac{\text{Area of shaded region}}{\text{Area of square}} = \frac{50\pi}{400} = \frac{\pi}{8}$$

$\xrightarrow{\div 50}$
 $\xrightarrow{\div 50}$

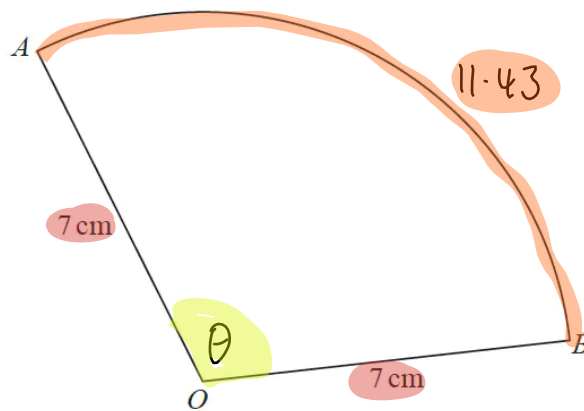
(Total for Question 7 is 4 marks)

[Edexcel, GCSE, Autumn 2018, Paper 1H, Q17]

Area of a Sector Questions from Past Papers

4

OAB is a sector of a circle with centre O and radius 7 cm.



The area of the sector is 40 cm^2

Calculate the perimeter of the sector.

Give your answer correct to 3 significant figures.

$$\text{Area of sector} = \frac{\theta}{360} \times 7^2 \times \pi = 40$$

$$\frac{\theta}{360} \times 49\pi = 40$$

$$\frac{\theta}{360} = \frac{40}{49\pi}$$

$$\theta = 360 \times \frac{40}{49\pi}$$

$$\theta = 93.544^\circ$$

$$\text{Circumference of circle} = 2 \times 7 \times \pi = 14\pi$$

$$\begin{aligned} \text{Arc length of sector} &= 14\pi \times \frac{93.544}{360} \\ &= 11.43 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Perimeter} &= 11.43 + 7 + 7 \\ &= 25.4285 \end{aligned}$$

$$25.4 \text{ cm}$$

(Total for Question 12 is 4 marks)

[Edexcel, GCSE, Summer 2019, Paper 2H, Q12]

Area of a Sector Questions from Past Papers

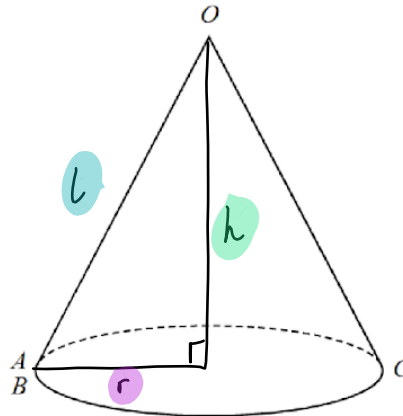
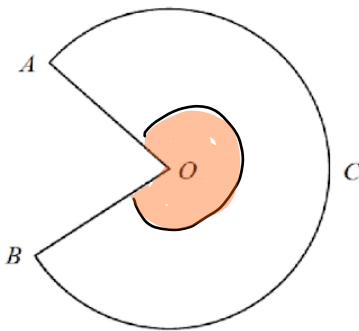
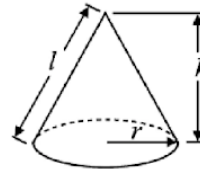
5

The diagram shows a sector $OACB$ of a circle with centre O . The point C is the midpoint of the arc AB .

The diagram also shows a hollow cone with vertex O . The cone is formed by joining OA and OB .

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Curved surface area of cone} = \pi r l$$



The cone has volume 56.8 cm^3 and height 3.6 cm .

Calculate the size of angle AOB of sector $OACB$. Give your answer correct to 3 significant figures. You must show all your working.

$$\frac{1}{3} \pi r^2 h = 56.8$$

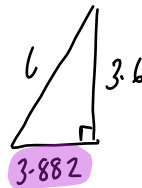
$$\frac{1}{3} \times \pi \times r^2 \times 3.6 = 56.8$$

$$1.2 \pi r^2 = 56.8$$

$$r^2 = \frac{56.8}{1.2\pi}$$

$$r = \sqrt{\text{ANS}}$$

$$r = 3.882$$



$$l^2 = 3.6^2 + 3.882^2$$

$$l^2 = 28.03$$

$$l = 5.29$$

$$\text{Curved surface area} = \pi \times 3.882 \times 5.29 = 64.57$$

$$\text{Area of circle} = \pi \times 3.882^2 = 47.34$$

$$\frac{47.34}{64.57} \times 360 = 263.96$$

264

(Total for Question 23 is 5 marks)

[Edexcel, GCSE, Autumn 2019, Paper 3H, Q23]

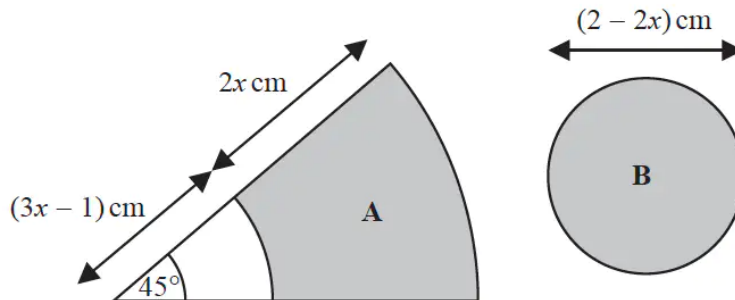
Area of a Sector Questions from Past Papers

6

The diagram shows two shaded shapes, A and B.

Shape A is formed by removing a sector of a circle with radius $(3x - 1)$ cm from a sector of the circle with radius $(5x - 1)$ cm.

Shape B is a circle of diameter $(2 - 2x)$ cm.



Shape B
Area of Circle
 $\pi(1-x)^2$
 $= (1-2x+x^2)\pi$

The area of shape A is equal to the area of shape B.

Find the value of x .

You must show all your working.

$$\frac{45}{360} \times \pi \times (5x-1)^2 = \frac{1}{8} (25x^2 - 10x + 1) \pi$$

Large Sector

$$\frac{45}{360} \times \pi \times (3x-1)^2 = \frac{1}{8} (9x^2 - 6x + 1) \pi$$

Small Sector

$$\text{Shape A} = \frac{1}{8} (25x^2 - 10x + 1) \pi - \frac{1}{8} (9x^2 - 6x + 1) \pi = \frac{1}{8} (16x^2 - 4x) \pi$$

$$\frac{1}{8} (16x^2 - 4x) \pi = (1 - 2x + x^2) \pi$$

$$\times 8 \downarrow \quad 16x^2 - 4x = 8(1 - 2x + x^2) \quad \times 8 \downarrow$$

$$16x^2 - 4x = 8 - 16x + 8x^2$$

$$8x^2 + 12x - 8 = 0$$

$$2x^2 + 3x - 2 = 0$$

$$(2x-1)(x+2) = 0$$

$$x = \frac{1}{2}$$

$$x = -2$$

(Total for Question 22 is 5 marks)

$x = \frac{1}{2}$ (can't be negative)

[Edexcel, GCSE, Autumn 2020, Paper 1H, Q22]

Area of a Sector Questions from Past Papers

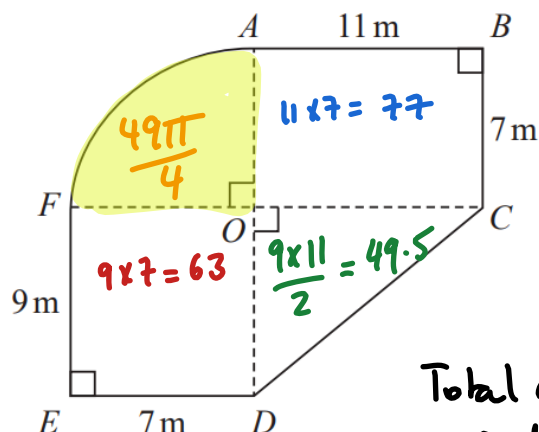
7

The diagram shows a plan of Jason's garden.

$ABCO$ and $DEFO$ are rectangles.

CDO is a right-angled triangle.

AFO is a sector of a circle with centre O and angle $AOF = 90^\circ$



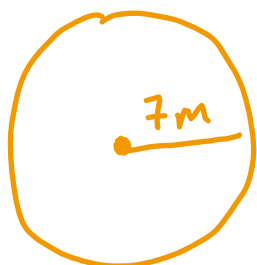
$$\begin{aligned} \text{Total area of garden} &= 63 + 77 + 49.5 + \frac{49\pi}{4} \\ &= 227.98 \text{ m}^2 \end{aligned}$$

Jason is going to cover his garden with grass seed.

Each bag of grass seed covers 14 m^2 of garden.

Each bag of grass seed costs £10.95

Work out how much it will cost Jason to buy all the bags of grass seed he needs.



$$\text{Area} = \pi \times 7^2 = 49\pi$$

$$\text{Area of sector} = \frac{49\pi}{4}$$

$$\frac{227.98}{14} = 16.28 \quad (17 \text{ bags})$$

$$17 \times 10.95 = \text{£} 186.15$$

(Total for Question 21 is 5 marks)

[Edexcel, GCSE, November 2022, Paper 2F, Q21]